

Answers — the faded set

1 • Why the diagonal, and why a mirror pair

Swapping $u \leftrightarrow v$ leaves the pair of equations unchanged. So the set of fixed points is symmetric under the swap: a crossing at (u, v) implies one at (v, u) . Crossings therefore come either on the diagonal, where the swap does nothing, or in mirror pairs. And there is *always* one on the diagonal because $x + x^{n+1} = \alpha$ has exactly one positive root — the left side increases from 0 without bound.

2 • The pair, classified

$$g_1 = \frac{4 \cdot 0.118 \cdot 8}{17} = 0.222, \quad g_2 = \frac{4 \cdot 2 \cdot 0.00164}{1.0002} = 0.013, \quad g_1 g_2 = 0.0029.$$

Stable: $\lambda = -1 \pm \sqrt{0.0029} = -0.946, -1.054$. A node, and a deep one — both eigenvalues near -1 , which is to say the two proteins relax almost independently, each at its own lifetime.

Why the product. A perturbation has to go round the loop — u changes dv/dt by g_2 , the change in v changes du/dt by g_1 — and what comes back is the product. The tiny factor is g_2 : at $u = 0.118$ the repressor u is far below its threshold, so promoter 2 does not feel it at all ($u^4 = 0.0002$). That is what a *committed* state looks like: one arm has lost its grip entirely.

3 • $n = 1$

At $u = v = 1$: $g_1 = g_2 = 1 \cdot 1 \cdot 1 / (1 + 1) = \frac{1}{2}$, so $g_1 g_2 = \frac{1}{4} < 1$. Stable node, $\lambda = -\frac{1}{2}, -\frac{3}{2}$.

One crossing in total. Both nullclines are hyperbolas, $u = 2/(1 + v)$ and $v = 2/(1 + u)$; neither has an inflection, so they meet once. Tuesday's handout proved the general statement: with $\beta = \gamma = 1$, $g_1 g_2 = uv / ((1 + u)(1 + v)) < 1$ at every point, so no crossing can ever be a saddle, and without a saddle there cannot be three crossings.

4 · Weaken one arm

Only the v -nullcline moves: $v = \alpha_2 / (1 + u^4)$ sinks toward the u -axis as α_2 falls; the u -nullcline is unchanged. The v -high crossing near $(0.12, 2)$ and the saddle approach each other and meet — a **saddle-node** — and the cell is left in the single remaining state, u high and v off: $(2.000, 0.059)$ at $\alpha_2 = 1$. From `stability_report`, three crossings survive down to $\alpha_2 \approx 1.4$ and only one remains by 1.3 ; the point has walked out of the wedge through one edge.

Same event as 39 μM IPTG, produced by construction instead of by inducer. In the paper it is **PIKE105**: the arm TetR represses ($P_{\text{LtetO-1}}$, driving *lacI*) was left too strong relative to the other, one state swallowed the other, and the plasmid sits monostable in its low, LacI-on state — the u -high state here. Whether you weaken one arm or strengthen the other is the same move in the wedge.